

LJUBLJANA BEZIGRAD, Slovenia

WMO: 140150

Lat: 46.0656N

Lon: 14.5122E

Elev: 978

StdP: 14.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	20.3	8.0	8.6	21.4	12.6	10.8	26.4	12.2	41.0	10.8	40.3	2.7	50	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	89.7	70.2	86.7	68.8	83.8	67.6	71.6	86.1	70.1	83.5	68.7	81.1	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	102.7	77.5	65.5	97.5	76.0	64.2	93.1	74.2	36.0	86.4	34.6	83.8	33.4	81.2	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	9.7	8.1	DB	11.8	94.2	5.0	3.4	8.2	96.7	5.3	98.7	2.5	100.6	-1.1	103.1	(4)
(5)			WB	10.8	74.4	4.8	2.1	7.3	75.9	4.5	77.1	1.8	78.3	-1.8	79.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	33.7	36.9	44.8	53.0	61.0	68.0	71.3	70.6	61.8	53.3	44.0	34.8	(6)
(7)		DBStd	14.84	7.07	6.94	6.99	6.49	5.79	6.24	5.46	5.66	5.87	6.83	7.68	7.41	(7)
(8)		HDD50	1830	505	368	186	44	2	0	0	0	0	45	207	473	(8)
(9)		HDD65	5021	970	787	627	360	149	41	12	14	131	364	631	935	(9)
(10)		CDD50	2874	0	1	24	135	344	540	661	640	353	148	26	2	(10)
(11)		CDD65	591	0	0	0	1	27	131	208	189	34	1	0	0	(11)
(12)		CDH74	4989	0	0	0	26	249	1120	1822	1566	202	4	0	0	(12)
(13)		CDH80	1609	0	0	0	1	34	346	641	557	30	0	0	0	(13)
(14)	Wind	WSAvg	3.1	2.7	3.1	3.8	3.8	3.7	3.4	3.3	3.0	2.8	2.9	2.7	2.5	(14)
(15)	Precipitation	PrecAvg	53.6	2.6	3.0	3.4	3.9	4.3	5.5	5.0	5.2	5.7	5.6	5.6	3.7	(15)
(16)		PrecMax	73.5	7.5	10.7	7.7	6.1	8.1	8.2	9.5	10.5	14.6	15.8	13.5	9.1	(16)
(17)		PrecMin	40.4	0.2	0.1	0.4	0.1	1.6	2.1	1.7	0.9	1.6	0.1	0.0	0.0	(17)
(18)		PrecStd	7.5	1.7	2.4	2.1	1.5	1.6	1.5	1.9	2.4	3.1	3.4	3.1	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	60.1	69.3	77.7	83.9	90.7	93.9	94.3	83.6	73.5	63.3	56.3	(19)
(20)			MCWB	47.4	46.7	52.7	59.0	65.9	70.6	71.3	71.1	67.4	62.6	55.2	51.6	(20)
(21)		2%	DB	49.8	53.9	64.2	72.6	79.8	87.2	89.8	89.6	79.5	69.6	59.3	52.0	(21)
(22)			MCWB	45.5	45.3	50.8	56.3	63.1	69.1	70.2	70.0	66.0	60.7	54.3	48.6	(22)
(23)		5%	DB	47.4	50.6	60.0	69.0	76.6	84.2	86.6	86.1	75.7	66.4	56.7	48.8	(23)
(24)			MCWB	43.9	44.3	48.9	54.8	61.6	67.5	68.9	68.7	63.8	58.8	53.0	45.9	(24)
(25)		10%	DB	44.8	47.4	56.2	65.3	73.3	80.9	83.6	82.5	72.1	63.3	54.5	45.3	(25)
(26)			MCWB	41.7	42.1	47.4	53.2	59.8	66.0	67.7	67.5	62.1	57.5	51.2	42.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	49.2	54.1	60.3	67.4	72.7	73.7	73.4	68.7	63.7	57.1	52.2	(27)
(28)			MCD B	51.6	54.9	65.6	74.3	81.3	87.3	89.1	89.8	80.2	70.8	60.7	55.3	(28)
(29)		2%	WB	46.5	47.0	51.9	57.6	64.6	70.6	71.7	71.4	66.7	61.9	55.3	49.1	(29)
(30)			MCD B	49.2	52.5	60.7	69.5	76.8	84.3	86.2	85.8	77.4	67.9	58.2	52.0	(30)
(31)		5%	WB	44.5	45.0	50.4	55.7	62.6	68.7	70.1	69.9	65.1	60.1	53.5	46.2	(31)
(32)			MCD B	47.1	50.2	57.8	66.4	74.2	81.5	83.7	82.8	73.8	65.1	56.6	48.8	(32)
(33)		10%	WB	42.1	42.8	48.8	54.0	60.9	66.8	68.5	68.4	63.3	58.4	51.6	43.0	(33)
(34)			MCD B	44.6	47.1	55.2	63.1	71.3	78.5	81.0	80.2	70.2	62.7	54.3	45.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	13.1	16.6	17.4	18.4	18.8	19.7	19.3	16.6	13.8	9.4	8.3	(35)
(36)			MCDBR	12.7	20.4	24.8	25.0	24.5	23.7	24.4	24.9	21.6	17.7	12.7	12.1	(36)
(37)			MCWBR	8.8	12.2	13.0	11.3	10.4	9.3	9.0	8.9	9.5	9.3	8.1	9.0	(37)
(38)		5% WB	MCDBR	10.3	18.0	20.5	21.8	22.3	22.3	22.5	22.5	19.7	14.8	10.3	10.6	(38)
(39)			MCWBR	7.5	11.4	11.3	10.2	9.9	9.6	8.9	8.9	9.1	8.4	7.3	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.353	0.400	0.450	0.449	0.457	0.453	0.447	0.425	0.400	0.359	0.324	(40)	
(41)		taud	2.397	2.315	2.212	2.099	2.147	2.175	2.196	2.225	2.250	2.294	2.360	2.441	(41)	
(42)		Ebn at Noon	243	257	259	256	261	259	258	254	248	236	226	231	(42)	
(43)		Edh at Noon	23	30	39	48	47	46	44	42	37	31	24	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	398	652	1020	1350	1662	1883	1897	1660	1163	751	415	333	(44)	
(45)		RadStd	59	109	144	163	165	126	125	183	153	88	63	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	N/A	N/A	+2.11	+1.58	+1.26	N/A	-184	+214	+162
(47) Regional (69 neighbors)		+0.88	N/A	N/A	+1.46	+0.83	+0.65	-144	-229	+171	+112

Nomenclature: See separate page